

Robel Geda

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EDUCATION

Princeton University

Ph.D. Astrophysics

Advisor: Professor Jenny E. Greene

Princeton, NJ

August 2021 - Present

Rutgers University

B.S. Astrophysics & B.A. Computer Science

Advisor: Professor Alyson Brooks

New Brunswick, NJ

August 2012 - January 2017

EMPLOYMENT

Space Telescope Science Institute (STScI)

Software Engineer & Research and Instrument Analyst

Developer on the James Webb Data Analysis Development Forum (JDADF) and Nancy Grace Roman Space Telescope (formerly WFIRST) mission. Help desk lead for the Nancy Grace Roman Space Telescope mission.

Baltimore, MD

June 2017 – July 2021

RESEARCH

Research Interests: Galaxy formation and evolution (observational surveys & simulations)

Star Formation Histories of Dwarf Galaxies

Dr. Akaxia Cruz & Dr. Jenny E. Greene

Studying the SFH of dwarf galaxies using high-resolution cosmological zoom-in simulations, characterizing their variability across timescales to connect SFH diversity to morphological and environmental properties.

Princeton University

August 2025 - Present

The Formation of Dwarf Galaxy Disks

Dr. Akaxia Cruz & Dr. Jenny E. Greene

Used high-resolution cosmological zoom-in simulations of isolated dwarf galaxies to show that rotation supported stellar disks built from gas spun up by high angular-momentum satellite mergers drive the secular size growth that transforms compact dwarfs into extended galaxies ($R_e > 2 \text{ kpc}$).

Princeton University

July 2024 – August 2025

DM Halo Density Peaks & Merger Trees

Dr. Romain Teyssier

Developed a new watershed algorithm using phase-space analysis to identify density peaks in cosmological simulations, along with a merger tree code that uses the boosted potential to track peaks across time-steps.

Princeton University

July 2023 – July 2024

The High-mass X-Ray Binary Luminosity Functions of Dwarf Galaxies

Dr. Andy D. Goulding & Dr. Jenny E. Greene

Calibrated the high-mass X-ray binary luminosity function in nearby dwarf galaxies using Chandra archival data.

Princeton University

August 2021 – June 2023

Evolution of Luminous Compact Blue Galaxies

Dr. Steven Crawford

Identification and characterization of luminous compact blue galaxies (LCBGs) in the HST Frontier Fields. Developed a new 2D Sérsic profile fitting code to remove bright sources for improved photometry.

STScI

February 2020 – July 2021

Near-Field Cosmology with Gas-Poor Dwarf Galaxies

Dr. Harry Ferguson

Test the efficacy of convolutional neural networks (CNN) in identifying diffuse dwarf galaxies within distances of $\sim 2\text{--}30 \text{ Mpc}$. Wrote a library that uses stellar isochrones to simulate images of diffuse dwarfs with realistic fluctuations.

STScI

November 2018 – July 2021

Massively Parallel Cosmological N-body Code

Dr. Alyson Brooks & Dr. Tom Quinn

Utilized CUDA and C++ to optimize Ewald GPU calculations and data transfers.

Rutgers University

May 2016 – August 2019

Rutgers Compilers Group

Dr. Eddy Z. Zhang

GPU memory optimization by distribution workloads using edge partitioning algorithms.

Rutgers University

December 2015 – June 2017

Galaxy Formation and Evolution

Dr. Alyson Brooks

Study of dark matter halo merger rates using N-body Simulations. Developed custom halo tracker + merger tree generator for Amiga Halo Finder.

Rutgers University

September 2014 – June 2017

Summer R.E.U

Dr. Robert Harmon

Developed GPU matrix multiplication software for use in light curve inversions using CUDA and FORTRAN.

Ohio Wesleyan University

May – July 2015

Spectroscopy of Superconductors

Dr. Girsh Blumberg

Project aimed to characterize the surface vibration modes of Bismuth Selenide using Raman scattering. Collected Raman spectroscopic data and developed software for analysis on large spectroscopic datasets.

Rutgers University

September 2014 - May 2015

FUNCTIONAL PROJECTS

The Merian Survey DR2

Dr. Shany Danieli, Dr. Lee Kelvin, & Dr. Jenny E. Greene

Setup and tested data reduction pipeline for Data Release II (DR2) for the Merian Survey. Adapted and integrated quality check tools from the LSST Science Pipelines. Observed for 3 nights.

Princeton University

July 2024 – Present

James Webb Data Analysis Tools

Dr. Susan Kassin & Dr. Harry Ferguson

Served as lead developer and maintainer for the Multi Object Spectroscopy Visualization Tool (MOSViz); deputy lead developer for CubeViz (IFU data visualization). Lead liaison and deputy maintainer for STScI Science Notebooks.

STScI

July 2017 – July 2021

WebbPSF

Dr. Marshall Perrin

Developed, implemented and maintained the optical components for the Nancy Grace Roman Space Telescope.

STScI

July 2017 – July 2021

Roman-Tools

Dr. Karoline Gilbert

Served as lead developer and maintainer of the Roman Tools repository which holds instructions and tutorials for Nancy Grace Roman Space Telescope software tools distributed by STScI for the science community.

STScI

July 2017 – July 2021

STIPS

Dr. Karoline Gilbert & Brian York

Deputy lead developer of the STIPS (Space Telescope Image Product Simulator) software.

STScI

July 2017 – July 2021

LEADERSHIP AND TEACHING

Service & Outreach

Princeton Graduate Student Government – Social Chair

United Astronomy Clubs of New Jersey: Public Observing – Volunteer Observer

Astro Justice Forum – Organizer

Princeton University Public Observing – Volunteer Observer

AstroPy Inclusion, Diversity, Empowerment – Ambassador

L.S.A.M.P Rutgers – Scholar & Ambassador

January 2026 – Present

May 2023 – May 2025

January 2022 – January 2023

Spring & Fall 2022

July 2020 – August 2021

August 2012 – January 2017

Teaching Experience

AST 203: The Universe – Teaching Assistant

Software Carpentry – Certified Instructor

Spring 2024

August 2020 – August 2021

SOFTWARE

GitHub ID: [robelgeda](#)

Current Projects:

- PetroFit [Developer & Maintainer]: Maintainer for Petrosian profiling and Sérsic fitting library. pyOpenSci peer-reviewed Astropy affiliated package (1 of 6 as of Spring 2026).

Selected Past Contributions:

- SpecUtils [Developer]: AstroPy affiliated library for manipulating and analyzing spectroscopic data.
- WebbPSF [Developer & Maintainer]: PSF simulator for JWST and the Nancy Grace Roman Space Telescope.
- STIPS [Developer & Maintainer]: Post pipeline image simulator for the Nancy Grace Roman Space Telescope.
- Jdaviz [Developer]: Interactive data visualization and analysis tools for astronomical observations.
- ChaNGa [Contributor]: SPH N-body code. Contributed CUDA kernels to compute Ewald boundary conditions.
- RAMSES [Contributor]: AMR N-body code. Contributed to GitHub infrastructure and GPU hackathons

CONFERENCES AND TALKS

2025 Galactic Frontiers II. Poster: Dwarf Galaxy Disk Formation.	Hanover, NH
2025 Dancing in the Dark When Galaxies Shape Galaxies. Talk: Dwarf Galaxy Disk Formation.	Sexten, Italy
2025 Center for Computational Astrophysics. Talk: Dwarf Galaxy Disk Formation.	NYC, NY
2024 Center for Computational Astrophysics. Talk: The Star Formation Histories of Isolated Dwarf Galaxies	NYC, NY
2024 Ramses User Meeting, Talk: Phase-Space Density Peak Finding and Merger Trees	NYC, NY
2021 N.S.B.P, Talk: PetroFit.	Virtual
2020 N.S.B.P, STScI outreach representative and exhibitor.	Virtual
2020 SACNAS AstroPy outreach representative and STScI exhibitor.	Virtual
2019 N.S.B.P, STScI outreach representative and exhibitor.	Providence, RI
2019 SciPy, STScI attendee.	Austin, TX
2016 AAS 227, Poster: Starspots on LO Pegasi, 2006-2015.	Kissimmee, FL
2015 Aresty Research Symposium, Poster: Raman Spectroscopy and Simultaneous Fits.	New Brunswick, NJ
2015 N.S.B.P, Poster: Comparison of Merging Dark Matter Halo Histories.	Baltimore, MD
2015 Astro Hack Week 2015, Collaborated on Automated Voigt Profile Fitter.	New York, NY

CERTIFICATIONS

2021 CITI Program, Responsible Conduct of Research.	Princeton, NJ
2020 Software Carpentry, Instructor Certification.	Baltimore, MD
2018 Advanced Python Training, Advanced Python Certification	Baltimore, MD
2014 Rutgers Laser Safety, Laser Safety Certification.	New Brunswick, NJ
2014 Rutgers Lab Safety, Lab Safety Certification.	New Brunswick, NJ

AWARDS

2021 STScI Bravo Award, Delivery of Data Analysis Tools, build D.	Baltimore, MD
2021 STScI Achievement Award, JDAT Notebooks.	Baltimore, MD
2020 STScI Achievement Award, Roman PDR-2 Team.	Baltimore, MD
2020 STScI Bravo Award, Delivery of Jupyter Notebooks, build C.	Baltimore, MD
2020 STScI Bravo Award, Delivery of Data Analysis Tools, build C.	Baltimore, MD
2020 STScI Bravo Award, Delivery of Data Analysis Tools, build B.	Baltimore, MD
2020 STScI Bravo Award, Major improvements to S.T.I.P.S.	Baltimore, MD
2018 STScI Bravo Award, Developing WFIRST (Roman) simulation and science planning tools.	Baltimore, MD

First and Second Author:

1. *Disk Formation and the Size-sSFR Relation of Dwarf Galaxies.*
Geda, R., Cruz A., Wright A.C., Greene J.E., Brooks, A., Quinn, T., Wadsley, J., & Keller, B.; 2025
arXiv e-prints (accepted The Astrophysical Journal)
2. *Constructing merger trees of density peaks using phase-space watershed segmentation algorithm.*
Geda, R. and Teyssier, R; 2025
Monthly Notices of the Royal Astronomical Society 537, 1
3. *The HMXB Luminosity Functions of Dwarf Galaxies.*
Geda, R., Goulding, A. D., Lehmer, B. D., Greene, J. E., & Kulkarni, A.; 2024
The Astrophysical Journal 965, 67
4. *PetroFit: A Python Package for Computing Petrosian Radii and Fitting Galaxy Light Profiles.*
Geda, R., Crawford S., Hunt L.R., Bershad, M.A., Tollerud, E.J., & Randriamampandry, S.M.; 2022
The Astronomical Journal 163, 202
5. *A Simple Yet Effective Balanced Edge Partition Model for Parallel Computing.*
Li, L., **Geda, R.**, Hayes, A.B., Chen, Y., Chaudhari, P., Zhang, E.Z., & Szegedy, M.; 2017
ACM SIGMETRICS 45, 1

Contributing Author:

1. *First Data Release of the Merian Survey: A Wide-field Imaging Survey of Dwarf Galaxies at $z \sim 0.06\text{--}0.10$*
Danieli, S.; et al. 2025 (including **Geda, R**)
arXiv e-prints
2. *Dwarf diversity in Λ CDM with baryons*
Cruz, A., Brooks, A., Lisanti, M., Peter, A.H.G., **Geda, R**, Quinn, T.R., Tremmel, M.; et al. 2025
arXiv e-prints
3. *STIPS: The Nancy Grace Roman Space Telescope Imaging Product Simulator*
Stips Development Team.; et al. 2024 (including **Geda, R**)
Publications of the Astronomical Society of the Pacific 136, 124502
4. *An Empirical Framework Characterizing the Metallicity and Star-formation History Dependence of X-Ray Binary Population Formation and Emission in Galaxies.*
Lehmer, Bret D.; et al. 2024 (including **Geda, R**)
The Astrophysical Journal 977, 189
5. *The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package.*
Astropy Collaboration; et al. 2022 (including **Geda, R**)
The Astrophysical Journal 935, 167